

Part II, Chapter 5: Entropic Economics and Cognitive Commons

5.1 Introduction: Reimagining Economics through Entropy

The attention economy, as critiqued in Chapter 1, thrives on the monopolization of cognitive bandwidth, transforming attention into a commodity to fuel storage and compute revenue streams. This extractive paradigm, rooted in entropic coercion, constrains semantic trajectories to maximize platform control, as formalized by the RSVP fields in Chapter 2. In contrast, this chapter proposes an entropic economics framework grounded in the Relativistic Scalar-Vector Plenum (RSVP), where attention is not a scarce resource to be harvested but a dynamic flow within a cognitive commons, governed by coherence (Φ), semantic directionality ($\mathbf{v}$), and entropic balance (S).

Drawing on Appendices E (Entropic Economics and Attention Flow) and C (Category-Theoretic Semantic Infrastructure), we formalize this economy using category-theoretic and sheaf-theoretic structures, ensuring rigorous modeling of decentralized, entropy-respecting interactions. By integrating futarchy-inspired governance and distributed architectures like HYDRA and TARTAN (Appendix D), we outline a cognitive commons that empowers agents to navigate semantic spaces freely, countering the centralized control of Big Tech.

5.2 Attention as a Scalar Commodity: A Category-Theoretic Perspective

In traditional economic models, attention is treated as a finite resource, with agents $i \in \mathcal{A}$ allocating attention $a_i(t)$ subject to a conservation principle (Appendix E.1):

$$\sum_{i \in \mathcal{A}} a_i(t) = A_{\text{total}},$$

where A_{total} is the total cognitive capacity of the system. However, this scalar view oversimplifies the dynamic interplay of attention within complex semantic networks. To address this, we model attention allocation within the symmetric monoidal ∞ -category $\mathcal{C}_{\text{RSVP}}$ (Appendix C.1), where agents are objects, and morphisms represent attention flows modulated by the vector field $\mathbf{v}$.

Formally, let $\mathcal{A} \hookrightarrow \mathcal{C}_{\text{RSVP}}$ be the subcategory of agents, with each $A_i \in \mathcal{A}$ equipped with a local attention state $a_i(t) : A_i \rightarrow \mathbb{R}_{\geq 0}$. The monoidal structure of $\mathcal{C}_{\text{RSVP}}$ allows attention to be composed via a tensor product:

$$a_{i,j}(t) = a_i(t) \otimes a_j(t),$$

representing joint attention allocation between agents i and j . The conservation principle is enforced by a global section over the base category $\mathcal{B}$, where $\pi : \mathcal{C}_{\text{RSVP}} \rightarrow \mathcal{B}$ (Appendix C.3) projects attention states onto contextual domains.

The dynamics of attention flow are governed by the vector field equation (Appendix A.2):

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla \Phi + \nu \nabla^2 \mathbf{v} + \mathbf{F}_{\text{ext}},$$

where $\mathbf{F}_{\text{ext}}$ encodes external algorithmic interventions (e.g., recommendation systems). In an entropy-respecting economy, $\mathbf{F}_{\text{ext}}$ is minimized, allowing $\mathbf{v}$ to align with natural semantic gradients driven by Φ .

5.3 Sheaf-Theoretic Entropic Economics

To model the distributed nature of a cognitive commons, we employ a sheaf-theoretic framework, which ensures local coherence while allowing global flexibility. Define a sheaf $\mathcal{F} : \mathcal{O}(\Omega)^{\text{op}} \rightarrow \mathcal{C}_{\text{RSVP}}$ on the topological space Ω , where $\mathcal{O}(\Omega)$ is the category of open sets, and $\mathcal{F}(U)$ assigns to each open set $U \subseteq \Omega$ the category of local agent states and attention allocations.

For an open cover $\{U_i\}$ of Ω , the sheaf condition ensures gluing compatibility:

$$\mathcal{F}(U) \cong \lim \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j) \right),$$

where the limit enforces consistency across overlapping domains. The entropy field S (Appendix A.3) is represented as a section of a subsheaf $\mathcal{S} \subset \mathcal{F}$, with evolution:

$$\frac{\partial S}{\partial t} + \nabla \cdot (\mathbf{v}S) = \kappa \nabla^2 S + \Gamma(\Phi, \mathbf{v}),$$

ensuring that local attention allocations contribute to global entropic balance. The source term $\Gamma(\Phi, \mathbf{v})$ accounts for information injections from external stimuli or agent interactions, preventing monopolistic collapse by diffusing uncertainty.

The sheaf $\mathcal{F}$ is fibered over $\mathcal{B}$, with fibers $\mathcal{F}_b$ (for $b \in \mathcal{B}$) encoding domain-specific economic interactions. Cross-domain attention flows are modeled via sheaf morphisms:

$$\phi : \mathcal{F}_b \rightarrow \mathcal{F}_{b'},$$

preserving the coherence potential Φ and ensuring that economic interactions respect the topological structure of Ω .

5.4 Futarchy and Decentralized Governance

To operationalize entropic economics, we draw on futarchy-inspired governance, where decision-making is driven by prediction markets that optimize collective utility. In RSVP, utility is defined per agent as (Appendix E.2):

$$U_i(t) = \Phi_i(t) - \lambda S_i(t),$$

where $\Phi_i(t)$ measures local coherence, and $\lambda S_i(t)$ penalizes entropic rigidity. The global objective is to maximize the total utility:

$$U_{\text{total}}(t) = \sum_{i \in \mathcal{A}} U_i(t),$$

subject to the attention conservation constraint $\sum_i a_i(t) = A_{\text{total}}$.

In a category-theoretic framework, futarchy is modeled as a functor:

$$F_{\text{fut}} : \mathcal{A} \rightarrow \mathcal{P},$$

where $\mathcal{P}$ is the category of prediction markets, with objects as market states and morphisms as bets or outcomes. The functor F_{fut} maps agent states to market predictions, ensuring that decisions align with the variational objective of Chapter 4:

$$\mathcal{F}[M_t, \Phi, S] = \mathbb{E}_{M_t}[\ln P(I_t | M_t)] + \beta S(M_t).$$

Sheaf-theoretically, market outcomes are sections of $\mathcal{F}$, with global coherence enforced by the gluing condition, preventing local optima from dominating the commons.

5.5 HYDRA and TARTAN in Economic Coordination

The distributed architectures HYDRA and TARTAN (Appendix D) facilitate the implementation of entropic economics. HYDRA nodes $N_i = (\Phi_i, \mathbf{v}_i, S_i)$ represent individual agents, with attention flows coupled via:

$$\mathbf{v}_i(t+1) = \sum_j A_{ij} \mathbf{v}_j(t) + f_i(t).$$

TARTAN acts as a functor:

$$T : \mathcal{C}_{\text{RSVP}}^{\text{local}} \rightarrow \mathcal{C}_{\text{RSVP}}^{\text{global}},$$

mapping local economic interactions to a globally coherent cognitive commons. This ensures sparse, entropy-respecting coordination, where agents optimize $U_i(t)$ without centralized control, countering the chokepoint capitalism critiqued previously.

5.6 The Cognitive Commons: Breaking Monopolistic Collapse

The cognitive commons is a decentralized ecosystem where agents freely allocate attention guided by RSVP dynamics, rather than external algorithmic attractors. Sheaf cohomology, specifically Čech cohomology (as referenced in prior discussions), quantifies obstructions to global coherence:

$$H^1(\Omega, \mathcal{S}) \neq 0 \implies \text{no global section,}$$

indicating potential monopolistic traps where local coherence (Φ_i) conflicts with global entropy (S). RSVP mitigates this by enforcing homotopy colimits (Appendix C.2) for semantic merging, ensuring that economic interactions remain modular and scalable.

For example, consider a social platform akin to LinkedIn, where only a fraction of job seekers secure placements, as noted in prior discussions. In an RSVP-based commons, attention allocation $a_i(t)$ is optimized via futarchy, with market signals derived from $\mathcal{F}$, ensuring equitable distribution of cognitive resources and preventing vanity-driven entropic collapse.

5.7 Conclusion: Toward a Modular Economic Paradigm

Entropic economics, grounded in RSVP’s field-theoretic and categorical framework, redefines attention as a dynamic flow within a cognitive commons. By leveraging sheaf-theoretic structures and distributed architectures, it counters the monopolistic tendencies of Big Tech, fostering a decentralized, entropy-respecting economy. The next chapter will explore the ethical implications of this paradigm, formalizing RSVP as a moral architecture for cognitive sovereignty.